

COURSE OUTLINE

1 Excel Formulas 101	<i>Syntax, reference types, errors, auditing, shortcuts, etc.</i>
2 Conditionals & Logical Operators	<i>IF, AND, OR, NOT, ISERROR, ISNUMBER, etc.</i>
3 Basic Statistical Functions	<i>MAX/MIN, RANK, RAND(), SUMIFS/COUNTIFS, SUMPRODUCT, etc.</i>
4 Lookup/Reference Functions	<i>VLOOKUP/HLOOKUP, INDEX/MATCH, OFFSET, etc.</i>
5 Text Formulas	<i>TEXT/VALUE, LEFT/MID/RIGHT, SEARCH, TRIM, LEN, etc.</i>
6 Date & Time Functions	<i>DATEVALUE, TODAY/NOW, DATEDIF, YEARFRAC, EOMONTH, etc.</i>
7 Formula-Based Formatting	<i>Creating, editing, and managing formula-driven formatting rules</i>
8 Array Formulas	<i>Vertical/Horizontal arrays, double unary, TRANSPOSE, etc.</i>
9 Extra Bonus Functions	<i>HYPERLINK, INDIRECT, WEBSERVICE, FILTERXML, etc.</i>

SETTING EXPECTATIONS

1 We'll be using Excel for **Windows/PC** (Excel 2013-2019)

- *What you see on your screen **will not always match** mine, especially if you're using an older version of Excel*

2 We'll focus on many of Excel's most **powerful, widely-used** functions

- *Excel's formula library includes nearly **500 functions**; we won't cover some of the more specialized categories (like Financial or Engineering functions), or those which require knowledge outside the scope of this course*

3 The goal of this course is to help you **master the building blocks**

- *The beauty of Excel formulas is that no matter how complex they get, they're ALL comprised of simple pieces. We'll start by mastering each individual component before combining them in more complex ways*

4 Feeling stuck? **We've got your back.**

- *If you have questions about the course material, feel free to post a question and we'll be happy to help*
- *For project-specific questions, recommend posting to the **answers.microsoft.com** community forum*



FORMULAS 101

All formulas start with an **equals sign**

Arguments are always surrounded by **parentheses**

Arguments are separated by **commas** in the US, but other regions may use different list separators (like **semi-colons**)

= **MATCH**(lookup_value, lookup_array, [match_type])

The **function name** tells Excel what type of operation you're about to perform (*Excel offers ~500 functions*)

Note: Function names aren't case-sensitive, and aren't always required; basic arithmetic and logical operations often don't need one:

- = A1 + B1
- = A1 / B1
- = A1 > B1
- = A1 = B1

These are **arguments**, which vary by function and provide Excel with the info needed to evaluate a result

Note: Not all arguments are required; optional arguments are surrounded by square brackets (*like [match_type] above*)

Most functions have at least one required argument, but some don't require any, like **ROW()**, **COLUMN()**, **TODAY()** or **NOW()**



PRO TIP:

=MATCH(A2,

MATCH(lookup_value, lookup_array, [match_type])

As you begin writing a formula, the **Function ScreenTips** box will guide you through each individual argument – this is an extremely helpful tool!

Reference types allow you to “recycle” formulas across multiple cells, without having to manually update your references (which would be completely impractical)

Cell references are **relative** by default (A1). This **allows the reference to change** as the formula is copied to new cells

The **\$** symbol is used to create **fixed** references. You can fix entire cells (\$A\$1) or just the column (\$A1) or row (A\$1), which **prevents references from changing** as the formula is copied to new cells



PRO TIP:

Mastering reference types is my #1 tip for working efficiently with formulas

Relative Column & Row

	A	B	C
1	A1		
2			
3			
4			C4

Relative Column, Fixed Row

	A	B	C
1	A\$1		
2			
3			
4			C\$1

Fixed Column, Relative Row

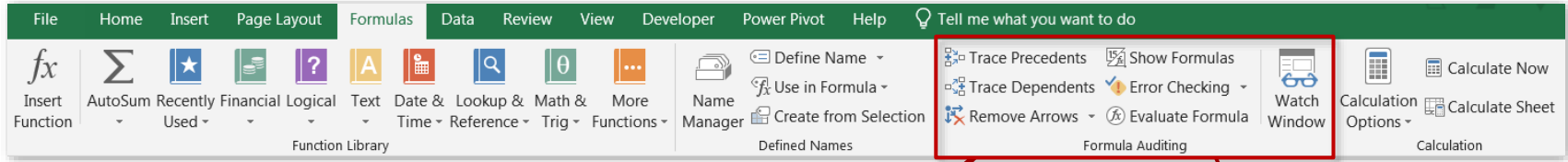
	A	B	C
1	\$A1		
2			
3			
4			\$A4

Fixed Column & Row

	A	B	C
1	\$A\$1		
2			
3			
4			\$A\$1

Common Errors

Error Type	What it means	How to fix it
#####	Column isn't wide enough to display values	<i>Drag or double-click column border to increase width, or right-click to set custom column width</i>
#NAME?	Excel does not recognize text in a formula	<i>Make sure that function names are correct, references are valid and spelled properly, and quotation marks and colons are in place</i>
#VALUE!	Formula has the wrong type of argument	<i>Check that your formula isn't trying to perform an arithmetic operation on text strings or cells formatted as text</i>
#DIV/0!	Formula is dividing by zero or an empty cell	<i>Check the value of your divisor; if 0 is correct, use an IF statement to display an alternate value if you choose</i>
#REF!	Formula refers to a cell that is not valid	<i>Make sure that you didn't move, delete, or replace cells that are referenced in your formula</i>
#N/A	Formula can't find a referenced value	<i>Check that all references and formula arguments evaluate properly (the most common cause is a lookup value with no match)</i>



Trace Precedents

Identifies cells which **affect** the value of the one selected

PROPERTY COST		Cash to Close:
Purchase Price	\$499,000	\$59,880
Tax Rate	\$9.50	
LOAN COST		\$3,021
Down Payment %	10%	
Interest Rate	4.50%	Monthly Expenses:
Term Length (yrs)	30	
Loan Amount	\$449,100	
Est. Closing Costs	\$9,980	
MONTHLY EXPENSES		
Mortgage Costs	\$2,276	
Property Tax	\$395	
Utilities	\$150	
Insurance	\$200	

Trace Dependents

Identifies cells which **are affected by** the value of the one selected

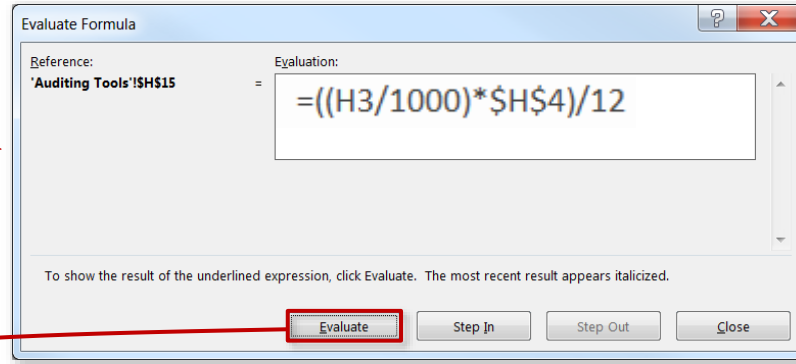
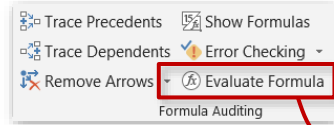
PROPERTY COST		Cash to Close:
Purchase Price	\$499,000	\$59,880
Tax Rate	\$9.50	
LOAN COST		\$3,021
Down Payment %	10%	
Interest Rate	4.50%	Monthly Expenses:
Term Length (yrs)	30	
Loan Amount	\$449,100	
Est. Closing Costs	\$9,980	
MONTHLY EXPENSES		
Mortgage Costs	\$2,276	
Property Tax	\$395	
Utilities	\$150	
Insurance	\$200	

Show Formulas

Temporarily displays all formulas in the worksheet as **text strings**

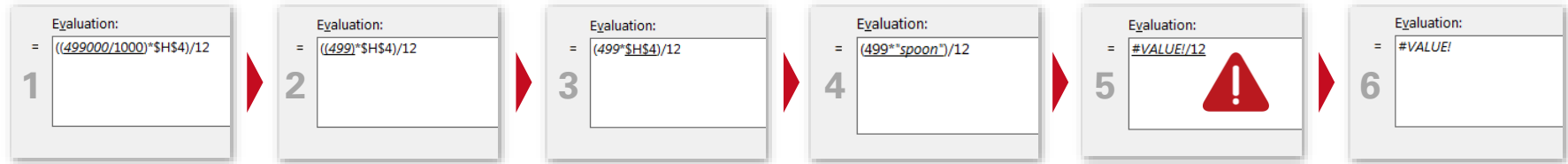
PROPERTY COST		Cash to Close:
Purchase Price	\$499,000	=(H3*H7)+H11
Tax Rate	\$9.50	
LOAN COST		Monthly Expenses:
Down Payment %	10%	
Interest Rate	4.50%	=SUM(\$H\$14:\$H\$17)
Term Length (yrs)	30	
Loan Amount	=H3*(1-H7)	
Est. Closing Costs	=H\$3*0.02	
MONTHLY EXPENSES		
Mortgage Costs	=1*PMT((\$H\$8/12),(\$H\$9*12),H\$10)	
Property Tax	=(H3/1000)*H\$4/12	
Utilities	\$150	
Insurance	\$200	

TIP: To select all cells containing formulas, use **Ctrl-G** to launch the Go-To menu, then select **Special > Formulas**



Evaluate Formula

Allows you to cycle through each individual calculation step within a formula, see how each component evaluates, and pinpoint the source of the error

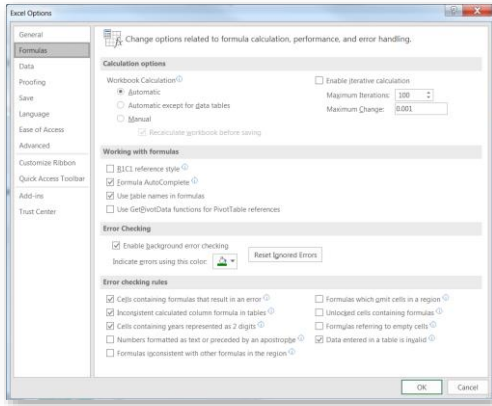
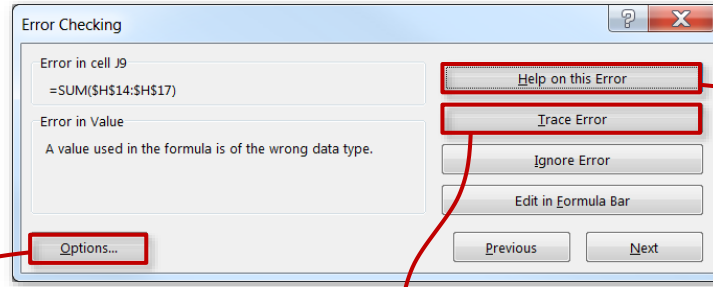


PRO TIP:

*Evaluate Formula is my **go-to tool** for breaking down complex or unfamiliar formulas*

Error Checking

Scans the sheet for errors and provides a summary with options to trace the source, ignore the error, modify your options, or link out to Microsoft support



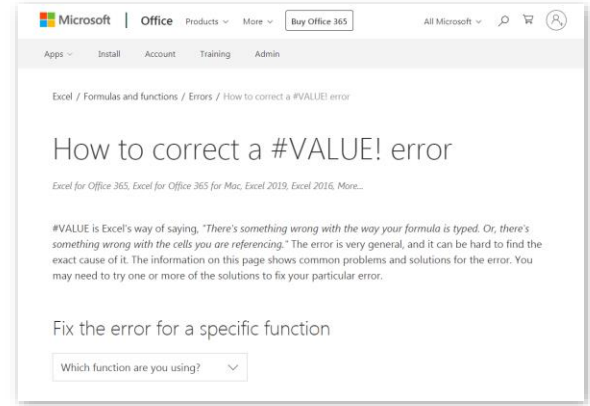
PROPERTY COST	
Purchase Price	\$499,000
Tax Rate	9.50

LOAN COST	
Down Payment %	10%
Interest Rate	4.50%
Term Length (yrs)	30
Loan Amount	\$449,100
Est. Closing Costs	\$9,980

MONTHLY EXPENSES	
Mortgage Costs	\$2,276
Property Tax	#VALUE!
Utilities	\$150
Insurance	\$200

Cash to Close:
\$59,880

Monthly Expenses:
#VALUE!



Ctrl + Arrow

- Jumps to the last cell in a data region, in the direction of the arrow

Ctrl + Shift + Arrow

- Selects to the last cell in a data region, in the direction of the arrow

Ctrl + Home/End

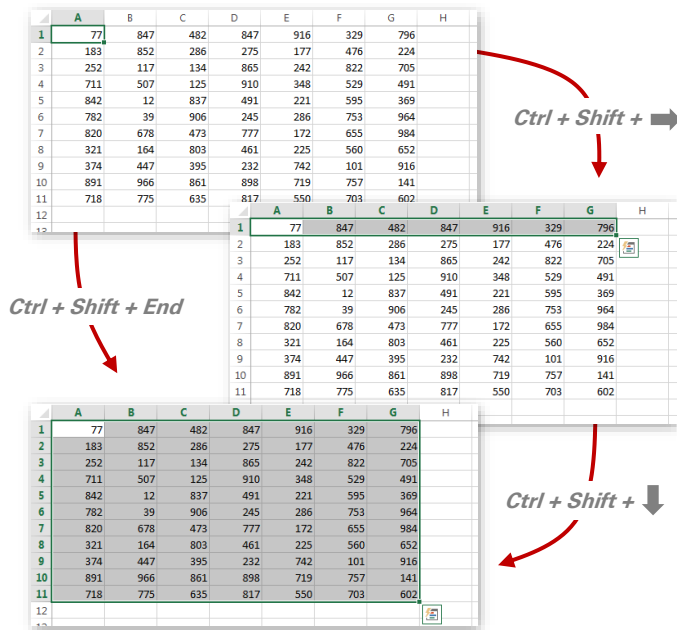
- Jumps to the **Home** (*top-left*) or **End** (*bottom-right*) cell in a region

Ctrl + .

- Jumps straight to **each corner** within a selected cell range

Ctrl + PgUp/PgDn

- Switches worksheet tabs, either to the **left** (*PgUp*) or **right** (*PgDn*)



F1

- Launches the **Excel help** pane (*default*)
- Links to the **Microsoft Support** website (*tool-specific*)

F2

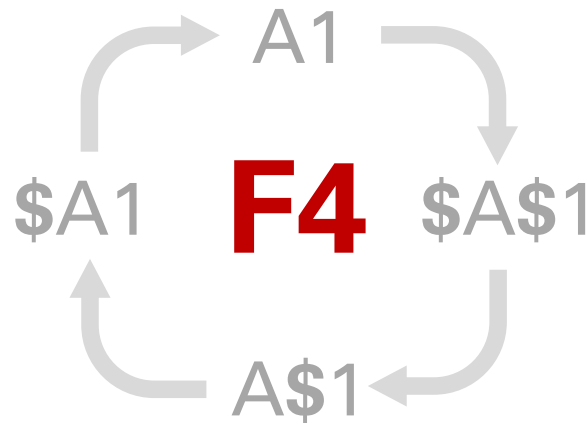
- Allows you to **edit** the active cell
- Highlights cells referenced by the active formula

F4

- Repeats the **last action** taken (*default*)
- Toggles **absolute/relative** cell references within a formula

F9

- Calculates all workbook formulas (*when in **manual** mode*)
- Evaluates **each function argument** within the formula bar



Common Mac Shortcuts

Mac Shortcut	Purpose	PC Equivalent
Command-T	<i>Cycles between cell reference types</i>	F4
Command-Y	<i>Repeats the last user action</i>	F4
Control-U	<i>Displays cell ranges tied to a given formula</i>	F2
Command-Arrow	<i>Jumps to the edge of a contiguous data array</i>	CTRL-ARROW
Command-Shift-Arrow	<i>Extends a selection to the edge of a data array</i>	CTRL-SHIFT-ARROW
Command-Fn-Up/Down	<i>Jumps between workbook tabs</i>	CTRL-PAGE UP/DOWN

Alt Key Tips

- Allow you to quickly access tools from ribbon menus and sub-menus using only the **keyboard** (*no clicks!*)
- Each keystroke takes you a layer deeper, until you land on the tool you need (**Note: Simply *press & release* the Alt key, instead of holding it down)**
- There are hundreds of combinations, so start by focusing on the tools that you use most frequently:

Some of my go-to key tips

Alt – H – V – V

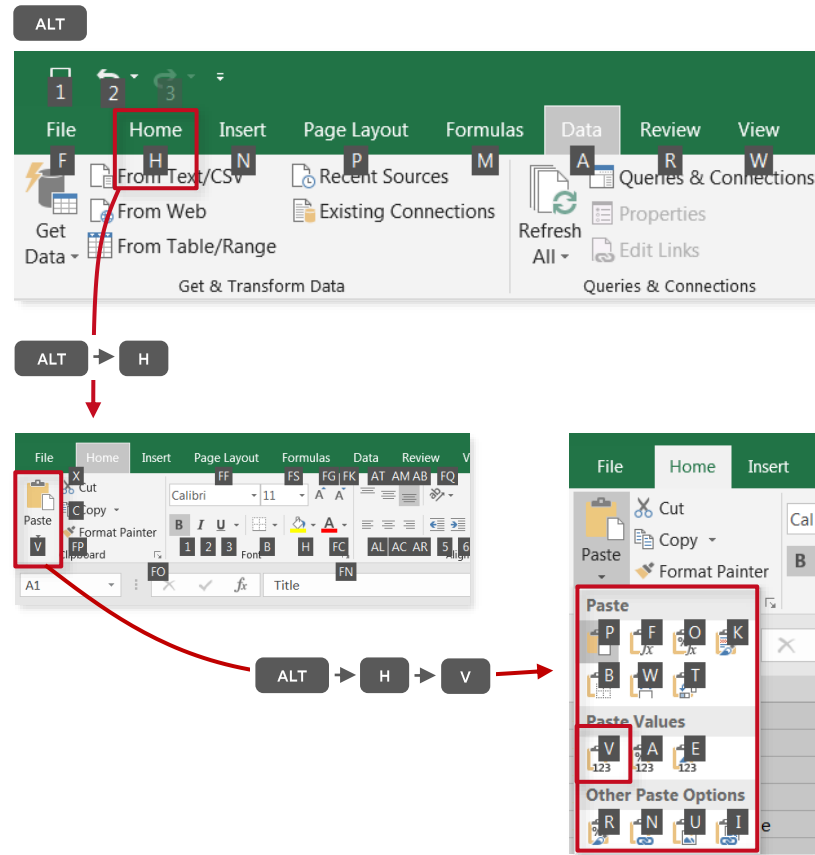
- Paste Special as Values

Alt – A – T

- Add or remove filters

Alt – M – V

- Evaluate Formula

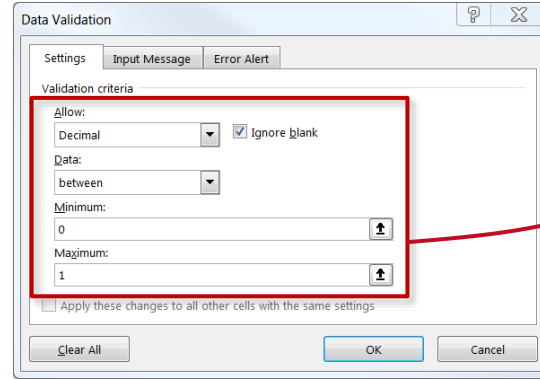
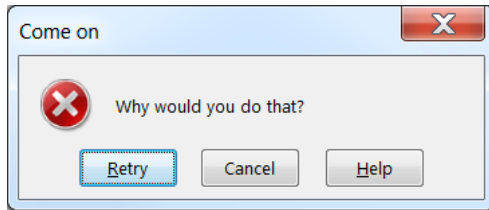


Data Validation

Restricts the values that a user can enter into a given cell, based on:

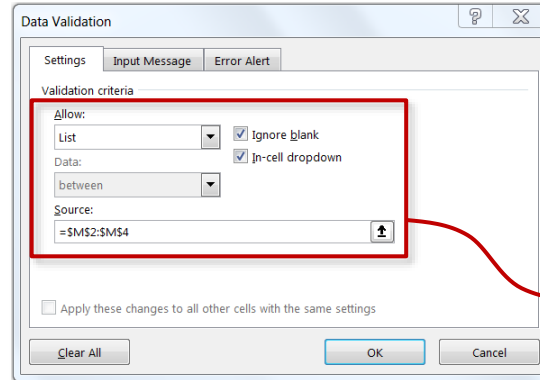
- **Number Type** (*Whole vs. Decimal*)
- **Value** (*Between, Less Than, Equal To, etc*)
- **List of Items** (*Based on cell range or manual list*)
- **Date/Time** (*Between, Less Than, Equal To, etc*)
- **Text Length** (*Between, Less Than, Equal To, etc*)
- **Custom** (*Formula-Driven*)

FUN FACT: You can customize your own **error alerts**!



LOAN COST	
Down Payment %	20%
Interest Rate	4.50%
Term Length (yrs)	30
Loan Amount	\$399,200
Est. Closing Costs	\$9,980

Decimal from 0 -1



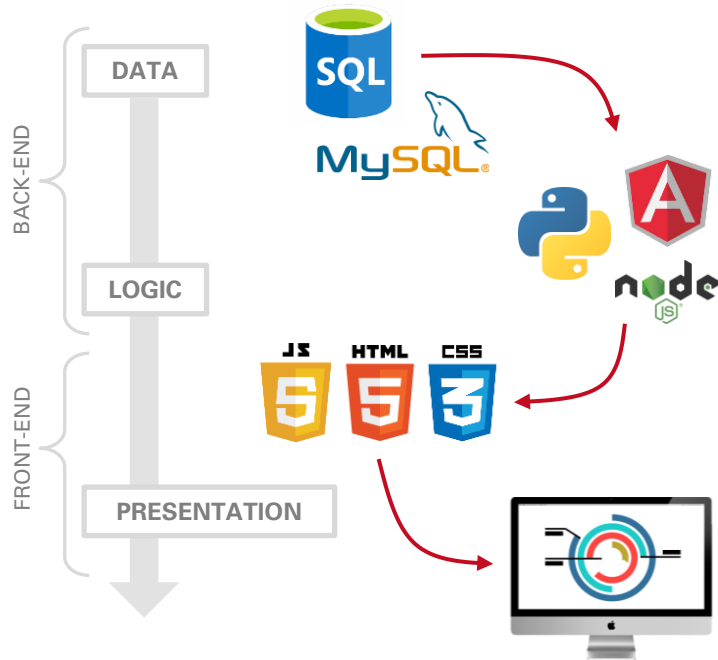
LOAN COST	
Down Payment %	20%
Interest Rate	4.50%
Term Length (yrs)	10
Loan Amount	10
Est. Closing Costs	15
	30

	M
1	Term Length:
2	10
3	15
4	30

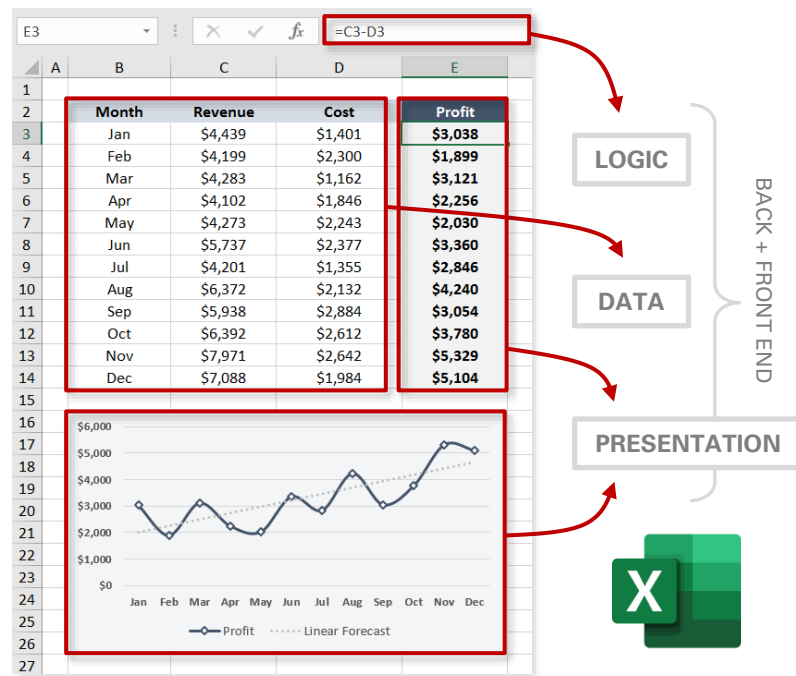
List of Items

Congrats, You're a Developer!

By definition, Excel is a **full-stack development** platform*; but rather than separating each layer of the process (*data, logic & presentation*), Excel mixes them all within the same user interface:



VS



**This is NOT a claim that Excel is always the right full-stack dev tool (or, in many cases, even a viable one). Rather, it's an effort to inspire users to think creatively about what Excel can do*

A horizontal banner with a dark green background. The background features faint, semi-transparent images of financial data, including a table with columns for dates, values, and percentages, and a line graph showing an upward trend. The text "CONDITIONAL STATEMENTS" is overlaid in large, white, bold, sans-serif capital letters.

CONDITIONAL STATEMENTS



All Conditional Statements in Excel are based on simple “IF/THEN” statements:

*-IF it's raining, **THEN** bring an umbrella*

*-IF it's sunny, **THEN** bring sunglasses*

*-IF it's sunny **AND** it's summer, skip work and go to the beach*

You're basically saying ***“Hey Excel, if this statement is true, do this. Otherwise, do something else.”***

=IF(logical_test, [Value if True], [Value if False])

Any test that results in either **TRUE** or **FALSE**

(i.e. $A1="Google"$, $B2<100$, etc)

Value returned if logical test is **TRUE**

Value returned if logical test is **FALSE**

	A	B	C	D
1	Location	Temp (F)	Precip (mm)	Freeze
2	A	75	0	No
3	B	18	0	Yes
4	C	86	0	No
5	D	80	2.3	No
6	E	28	1.2	Yes
7	F	68	0.5	No
8	G	26	0	Yes

= IF(B2<=0,"Yes","No")

In this case we're categorizing the Freeze column as "Yes" if the temperature is equal to or below 32, otherwise "No"

By using **Nested IF Statements**, you can include multiple logical tests within a single formula:

	A	B	C	D	E
1	Location	Temp (F)	Precip (mm)	Freeze	Climate
2	A	75	0	No	Mild
3	B	18	0	Yes	Cold
4	C	86	0	No	Hot
5	D	80	2.3	No	Mild
6	E	28	1.2	Yes	Cold
7	F	68	0.5	No	Mild
8	G	26	0	Yes	Cold

→ = IF(B2<40,"COLD",IF(B2>80,"HOT","MILD"))

*If temp<40, climate = "Cold", if temp>80,
climate = "Hot", otherwise climate = "Mild"*

Excel's **AND** and **OR** statements allow you to include multiple logical tests at once:

	A	B	C	D	E	F	G
1	Location	Temp (F)	Precip (mm)	Freeze	Climate	Precip Type	Conditions
2	A	75	0	No	Mild	None	Dry
3	B	18	0	Yes	Cold	None	Dry
4	C	86	0	No	Hot	None	Dry
5	D	80	2.3	No	Mild	Rain	Wet
6	E	28	1.2	Yes	Cold	Snow	Wet
7	F	68	0.5	No	Mild	Rain	Wet
8	G	26	0	Yes	Cold	None	Dry

=IF(OR(F2="Rain",F2="Snow"),"Wet","Dry")

Here we're categorizing conditions as "Wet" if the precipitation type equals "rain" OR "snow", otherwise Conditions = "Dry"

=IF(AND(D2="Yes",C2>0),"Snow",IF(AND(D2="No",C2>0),"Rain","None"))

If the temp is below freezing AND the amount of precipitation > 0, then Precip Type = "Snow", if the temp is above freezing AND the amount of precipitation >0, then Precip Type = "Rain", otherwise Precip Type = "None"



PRO TIP:

When writing nested functions, copy/paste repetitive pieces and tweak individual elements to save time (rather than starting from scratch)

If you want to evaluate a case where a logical statement is *not* true, you can use either the **NOT** statement or a “<>” operator

	A	B	C	D	E	F	G
1	Location	Temp (F)	Precip (mm)	Freeze	Climate	Precip Type	Conditions
2	A	75	0	No	Mild	None	Dry
3	B	18	0	Yes	Cold	None	Dry
4	C	86	0	No	Hot	None	Dry
5	D	80	2.3	No	Mild	Rain	Wet
6	E	28	1.2	Yes	Cold	Snow	Wet
7	F	68	0.5	No	Mild	Rain	Wet
8	G	26	0	Yes	Cold	None	Dry

=IF(NOT(C2=0),"Wet","Dry")

=IF(C2<>0,"Wet","Dry")

In both of these examples, we're defining Conditions = "Wet" if the amount of precipitation is NOT equal to 0

The **IFERROR** statement is an excellent tool to eliminate annoying error messages (**#N/A**, **#DIV/0!**, **#REF!**, *etc.*), which is particularly useful for front-end formatting

=IFERROR(value, value_if_error)



Formula or value that may or may not result in an error



Value returned in the case of an error



PRO TIP:

*If you're writing a formula that may trigger an error (i.e. a **VLOOKUP** where not all values have a match), **WRITE THE FULL FORMULA FIRST** then wrap it in an **IFERROR** statement*

Excel offers a number of different **IS** formulas, each of which checks whether a certain condition is true:

ISBLANK = Checks whether the reference cell or value is blank

ISNUMBER = Checks whether the reference cell or value is numerical

ISTEXT = Checks whether the reference cell or value is a text string

ISERROR = Checks whether the reference cell or value returns an error

ISEVEN = Checks whether the reference cell or value is even

ISODD = Checks whether the reference cell or value is odd

ISLOGICAL = Checks whether the reference cell or value is a logical operator

ISFORMULA = Checks whether the reference cell or value is a formula



COMMON STATS FUNCTIONS

The **Count, Average, Median, Mode, Max/Min, Percentile** and **Standard Deviation/Variance** functions are used to perform basic calculations on a data array

	A	B	C	D
1	Value			
2	90	Sample Size	19	=COUNT(A2:A20)
3	13			
4	22	Average:	51.47	=AVERAGE(A2:A20)
5	98			
6	61	Median:	54	=MEDIAN(A2:A20)
7	68			
8	50	Mode:	22	=MODE(A2:A20)
9	91			
10	16	Max:	98	=MAX(A2:A20)
11	23			
12	60	Min:	13	=MIN(A2:A20)
13	22			
14	56	25th Percentile	23	=PERCENTILE(A2:A20,.25)
15	54			
16	87	75th Percentile	68	=PERCENTILE(A2:A20,.75)
17	33			
18	68	Standard Deviation	28	=STDEV(A2:A20)
19	45			
20	21	Variance	767	=VAR(A2:A20)
21				

	A
1	Value
2	90
3	13
4	22
5	98
6	61
7	68
8	50

RANK(A2,A2:A8) = 2

RANK(A3,A2:A8) = 7 (*lowest*)

RANK(A4,A2:A8) = 6

RANK(A5,A2:A8) = 1 (*highest*)

RANK(A6,A2:A8) = 4

RANK(A7,A2:A8) = 3

RANK(A8,A2:A8) = 5

The **RANK** function returns the rank of a particular number among a list of values

The **SMALL/LARGE** functions return the n th smallest/largest values within an array

	A
1	Value
2	90
3	13
4	22
5	98
6	61
7	68
8	50

LARGE(A2:A8,2) = 90

(the 2nd largest number in the array is 90)

SMALL(A2:A8,3) = 50

(the 3rd smallest number in the array is 50)

	A	B
1	Value	Percent Rank
2	2,717	18%
3	3,485	24%
4	5,202	76%
5	3,612	29%
6	4,432	59%
7	2,699	12%
8	4,585	65%
9	6,003	94%
10	4,820	71%
11	2,550	6%
12	5,795	88%
13	4,240	41%
14	6,827	100%
15	4,359	53%
16	2,320	0%
17	5,775	82%
18	4,241	47%
19	3,966	35%

PERCENTRANK returns the rank of a value as a percentage of a given array or dataset

=PERCENTRANK(array, x)

What range of data
are you looking at?

Which **value** within the
range are you looking at?

PERCENTRANK(\$A\$2:\$A\$19, A14) = 100% (highest)

PERCENTRANK(\$A\$2:\$A\$19, A16) = 0% (lowest)

RAND() and **RANDBETWEEN** act like random number generators in Excel:

	A	B	C	D	E
1	0.5173	0.4091	0.7560	0.9012	0.2167
2	0.0906	0.2317	0.0906	0.5856	0.8646
3	0.1544	0.8240	0.4279	0.8782	0.7795
4	0.0097	0.0872	0.7740	0.9137	0.7815
5	0.2089	0.7028	0.0449	0.8173	0.9983
6	0.0761	0.4388	0.4056	0.5639	0.0668

The **RAND()** function returns a random value between 0 and 1 (to 15 digits)

The **RANDBETWEEN** function returns an integer between two values that you specify

	A	B	C	D	E
1	83	23	64	62	92
2	59	45	40	50	91
3	24	37	70	30	32
4	54	85	69	55	3
5	73	12	36	53	2
6	29	72	68	59	99

=RANDBETWEEN(0,100)

The **SUMPRODUCT** formula multiplies corresponding cells from multiple arrays and returns the sum of the products (*Note: all arrays must have the same dimensions*)

=SUMPRODUCT(array1, array2 ... array_N)

Example: Total Revenue

	A	B	C	D
1	Product	Quantity	Price	Revenue
2	Apple	2	\$0.50	\$1.00
3	Banana	4	\$1.00	\$4.00
4	Orange	3	\$0.80	\$2.40
5	Total			\$7.40

*Without using SUMPRODUCT, you could multiply quantity*price in each row and sum the products*

	A	B	C	D
1	Product	Quantity	Price	Revenue
2	Apple	2	\$0.50	
3	Banana	4	\$1.00	
4	Orange	3	\$0.80	
5	Total			\$7.40

SUMPRODUCT(B2:B4,C2:C4) = \$7.40

SUMPRODUCT is often used with filters to calculate products *only* for rows that meet certain criteria:

	A	B	C	D
1	Store	Product	Quantity	Price
2	Stop & Shop	Apple	2	\$0.50
3	Shaws	Banana	4	\$1.00
4	Market Basket	Banana	3	\$1.00
5	Trader Joe's	Pineapple	8	\$2.50
6	Stop & Shop	Orange	2	\$0.80
7	Shaws	Apple	1	\$0.50
8	Market Basket	Apple	5	\$0.50
9	Trader Joe's	Banana	6	\$1.00
10	Market Basket	Pineapple	3	\$2.50
11	Trader Joe's	Orange	8	\$0.80
12	Stop & Shop	Pineapple	3	\$2.50
13	Shaws	Pineapple	5	\$2.50
14	Stop & Shop	Banana	2	\$1.00
15	Shaws	Orange	6	\$0.80
16	Market Basket	Orange	7	\$0.80
17	Trader Joe's	Apple	3	\$0.50

Quantity of goods sold at Shaws:

SUMPRODUCT((A2:A17="Shaws")*C2:C17) = 16

Total revenue from Shaws:

SUMPRODUCT((A2:A17="Shaws")*C2:C17*D2:D17) = \$21.80

Revenue from apples sold at Shaws:

SUMPRODUCT((A2:A17="Shaws")*(B2:B17="Apple")*C2:C17*D2:D17) = \$0.50



PRO TIP:

When you add filters to a SUMPRODUCT, you need to change the commas to multiplication signs

Great, but how does it *really* work?

SUMPRODUCT((A2:A17="Shaws")*(B2:B17="Apple")*C2:C17*D2:D17) = \$0.50

	A	B	C	D
1	Store	Product	Quantity	Price
2	Stop & Shop	Apple	2	\$0.50
3	Shaws	Banana	4	\$1.00
4	Market Basket	Banana	3	\$1.00
5	Trader Joe's	Pineapple	8	\$2.50
6	Stop & Shop	Orange	2	\$0.80
7	Shaws	Apple	1	\$0.50
8	Market Basket	Apple	5	\$0.50
9	Trader Joe's	Banana	6	\$1.00
10	Market Basket	Pineapple	3	\$2.50
11	Trader Joe's	Orange	8	\$0.80
12	Stop & Shop	Pineapple	3	\$2.50
13	Shaws	Pineapple	5	\$2.50
14	Stop & Shop	Banana	2	\$1.00
15	Shaws	Orange	6	\$0.80
16	Market Basket	Orange	7	\$0.80
17	Trader Joe's	Apple	3	\$0.50

What YOU see

What EXCEL sees

*When you apply a condition or filter to a column, Excel translates those cells as **0's** (if false) and **1's** (if true)*

*If you multiply all four columns, **ONLY ROWS THAT SATISFY ALL CONDITIONS WILL PRODUCE A NON-ZERO SUM***

	A	B	C	D
1	Store	Product	Quantity	Price
2	0	1	2	\$0.50
3	1	0	4	\$1.00
4	0	0	3	\$1.00
5	0	0	8	\$2.50
6	0	0	2	\$0.80
7	1	1	1	\$0.50
8	0	1	5	\$0.50
9	0	0	6	\$1.00
10	0	0	3	\$2.50
11	0	0	8	\$0.80
12	0	0	3	\$2.50
13	1	0	5	\$2.50
14	0	0	2	\$1.00
15	1	0	6	\$0.80
16	0	0	7	\$0.80
17	0	1	3	\$0.50

The **COUNTIF**, **SUMIF**, and **AVERAGEIF** formulas calculate a sum, count, or average based on specific criteria

	A	B
1	Name	Age
2	George	90
3	Maria	13
4	Ryan	22
5	Tim	98
6	George	61
7	Tim	68
8	Tim	50
9	Maria	91
10	George	16
11	Maria	23
12	Tim	60
13	Ryan	22
14	Maria	56
15	George	54
16	George	87
17	Ryan	33
18	Ryan	68
19	Ryan	45
20	George	21

=COUNTIF(range, criteria)

=SUMIF(range, criteria, sum_range)

=AVERAGEIF(range, criteria, average_range)

Which cells need to
match your criteria?

Under what condition
do I want to sum, count,
or average?

Where are the values
that I want to sum or
average?

COUNTIF(B2:B20,22) = **2**

SUMIF(A2:A20,"Ryan",B2:B20) = **190**

SUMIF(A2:A20,"<>Tim",B2:B20) = **702**

AVERAGEIF(A2:A20,"Maria",B2:B20) = **45.75**

COUNTIFS, SUMIFS, and AVERAGEIFS are used when you want to evaluate a count, sum, or average based on *multiple* conditions or criteria

=COUNTIFS(criteria_range1, criteria1, criteria_range2 , criteria2...)

=SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2 , criteria2...)

=AVERAGEIFS(average_range, criteria_range1, criteria1, criteria_range2 , criteria2...)

	A	B	C	D
1	Month	Tactic	Campaign	Clicks
2	Jan	Search	Google	166
3	Jan	Search	MSN	263
4	Jan	Display	Contextual	289
5	Jan	Display	Retargeting	137
6	Feb	Search	Google	124
7	Feb	Search	MSN	311
8	Feb	Display	Contextual	350
9	Feb	Display	Retargeting	384
10	Mar	Search	Google	168
11	Mar	Search	MSN	358
12	Mar	Display	Contextual	347
13	Mar	Display	Retargeting	390

COUNTIFS(B2:B13,"Search", D2:D13,">200") = 3

SUMIFS(D2:D13, A2:A13,"Feb",B2:B13,"Display") = 734

AVERAGEIFS(D2:D13, A2:A13,"Jan",C2:C13,"MSN") = 263



PRO TIP:

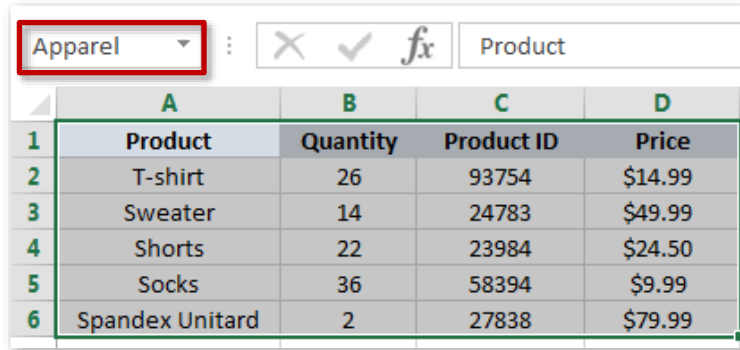
If you use < or >, you need to add quotation marks as you would with text (i.e. ">200")

A horizontal banner with a dark green background. It features a faint, semi-transparent image of an Excel spreadsheet. The spreadsheet contains various data points, including dates (e.g., 11/2/2014, 11/9/2014, 11/16/2014), numerical values, and percentages. There are also small bar charts and line graphs visible within the spreadsheet grid.

LOOKUP & REFERENCE FUNCTIONS

Using **Named Arrays** can simplify a lookup function if you use the same data array in multiple formulas

For example, if you name the array from A1:D6 "Apparel"...



The screenshot shows an Excel interface. At the top, a dropdown menu is set to 'Apparel', with a red box highlighting it. To its right are icons for deleting (X), confirming (checkmark), and inserting a function (fx). Below this is a table with columns A through D. Column A is 'Product', B is 'Quantity', C is 'Product ID', and D is 'Price'. The rows contain data for T-shirt, Sweater, Shorts, Socks, and Spandex Unitard.

	A	B	C	D
1	Product	Quantity	Product ID	Price
2	T-shirt	26	93754	\$14.99
3	Sweater	14	24783	\$49.99
4	Shorts	22	23984	\$24.50
5	Socks	36	58394	\$9.99
6	Spandex Unitard	2	27838	\$79.99

...you can write your vlookup formula in either of the following ways:

=VLOOKUP(A1,\$A\$1:\$D\$6,2)

=VLOOKUP(A1,Apparel,2)

Let's take a look at one of Excel's most common reference functions – **VLOOKUP**:

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

This is the **value** that you are trying to match in the table array

This is **where** you are looking for the lookup value

Which column contains the data you're looking for?

Are you trying to match the **exact** lookup value (0), or something similar (1)?

	A	B	C	D
1	Product	Quantity	Product ID	Price
2	T-shirt	26	93754	\$14.99
3	Sweater	14	24783	\$49.99
4	Shorts	22	23984	\$24.50
5	Socks	36	58394	\$9.99
6	Spandex Unitard	2	27838	\$79.99

D2=VLOOKUP(A2, \$G\$1:\$H\$5, 2, 0)

G	H
Product	Price
Shorts	\$24.50
Sweater	\$49.99
Spandex Unitard	\$79.99
T-shirt	\$14.99
Socks	\$9.99

To populate the Price in column D, we look up the name of the product in the data array from G1:H5 and return the value from the 2nd column over

Use **HLOOKUP** if your table array is transposed (variables headers listed in rows)

=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

This is the **value** that you are trying to match in the table array

This is **where** you are looking for the lookup value

Which row contains the data you're looking for?

Are you trying to match the **exact** lookup value (0), or something similar (1)?

	A	B	C	D
1	Product	Quantity	Product ID	Price
2	T-shirt	26	93754	\$14.99
3	Sweater	14	24783	\$49.99
4	Shorts	22	23984	\$24.50
5	Socks	36	58394	\$9.99
6	Spandex Unitard	2	27838	\$79.99

D2=HLOOKUP(A2, \$H\$1:\$L\$2, 2, 0)

With an HLOOKUP, we search for the product name in F1:J2 and return the value from the 2nd row down

G	H	I	J	K	L
Product	Shorts	T-shirt	Sweater	Spandex Unitard	Socks
Price	\$24.50	\$14.99	\$49.99	\$79.99	\$9.99

There are **two key rules** that constrain **VLOOKUP** and **HLOOKUP** formulas:



1. The lookup value must be in the **first column** of a VLOOKUP table array or the **first row** of a HLOOKUP table array
2. Excel will always return the value from the **top most row** or **left most column** of a table array when multiple instances of the lookup value are present



PRO TIP:

Avoid breaking Law #2 by identifying a “Key” that is common to both datasets and is unique for every row (NOTE: Keys often take the form of a concatenation of multiple fields)

The **ROW** function returns the row number of a given *reference*, while the **ROWS** function returns the number of rows in a given *array* or *array formula*

=ROW([reference])

=ROWS(array)

This example uses an array, which is why it includes the fancy {} signs – more on that in the ARRAY functions section

ROW(C10) = 10

ROWS(A10:D15) = 6

ROWS({1,2,3;4,5,6}) = 2

The **COLUMN** function returns the column number of a given *reference*, while the **COLUMNS** function returns the number of columns in a given *array* or *array formula*

=COLUMN([reference])

=COLUMNS(array)



PRO TIP:

Leave the cell reference out and just write ROW() or COLUMN() to return the row or column number of the cell in which the formula is written

COLUMN(C10) = 3

COLUMNS(A10:D15) = 4

COLUMNS({1,2,3;4,5,6}) = 3

The **INDEX** function returns the *value* of a specific cell within an array

=INDEX(array, row_num, column_num)

What range of cells
are you looking at?

How many rows
down is the value
you want?

How many columns
over is the value you
want?

	A	B	C
1	Tools	Price	Inventory
2	Hammer	\$5.00	55
3	Screw Driver	\$2.50	66
4	Pliers	\$3.34	333
5	Wrench set	\$10.00	234
6	Chain Saw	\$55.48	23
7	Tool Box	\$19.99	5
8	Level	\$2.25	7

INDEX(\$A\$1:\$C\$5, 5, 3) = 234

*In this case we're telling Excel to find the value of a cell somewhere within the array of A1:C5. Starting from the upper left, we move down to the **5th row** and right to the **3rd column**, to return the value of **234***

The **MATCH** function returns the *position* of a specific value within a column or row

=MATCH(lookup_value, lookup_array, [match_type])

What value are you trying to find the position of?

In which row or column are you looking? (**must be a 1-dimensional array**)

Are you looking for the exact value (0), or anything close?

1: Find largest value $\leq$ or = lookup_value

0: Find exact lookup_value

-1: Find smallest value $\geq$ or = lookup_value

	A	B
1	Tools	Price
2	Hammer	\$5.00
3	Screw Driver	\$2.50
4	Pliers	\$3.34
5	Wrench set	\$10.00

MATCH("Pliers", \$A\$1:\$A\$5, 0) = 4

	A	B	C
1	Tools	Price	Inventory
2	Hammer	\$5.00	55
3	Screw Driver	\$2.50	66
4	Pliers	\$3.34	333

MATCH(66, \$A\$3:\$C\$3, 0) = 3

*Matching the word "Pliers" in column A, we find it in the **4th** row. Matching the number 66 in row 3, we find it in the **3rd** column*

INDEX and **MATCH** are commonly used in tandem to act like a LOOKUP function; the only difference is that **INDEX/MATCH** can find values in any column or row in an array

Example: Price Checker

	A	B	C	D
1		Small	Medium	Large
2	Sweater	\$10	\$12	\$15
3	Jacket	\$30	\$35	\$40
4	Pants	\$25	\$30	\$35
5				
6	Product:	Pants		
8	Size:	Medium		
10	PRICE:	?		
11				

In this example, we want to populate the price of a given product and size in cell B10 by returning a particular value within the array B2:D4

B10=INDEX(B2:D4, MATCH(B6,A2:A4,0), MATCH(B8,B1:D1,0))

The number of rows down to index depends on what product I'm looking for, so we use a MATCH function and search for the value in cell B6 (in this case "Pants")

The number of columns over to index depends on what size I'm looking for, so we use a MATCH function and search for the value in cell B8 (in this case, "Medium")

Considering the output of each MATCH function, the formula is just a simple INDEX:

B10 = INDEX(B2:D4, 3, 2) = \$30

XLOOKUP can retrieve values from a table or range by matching a lookup value, and offers more flexibility than **VLOOKUP**, **HLOOKUP**, or **INDEX & MATCH** formulas

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])



Which **value** are you looking to match?



Where are you trying to find a match for your lookup value?



Where are the **values** you want to retrieve?



What if the lookup value **isn't found** in the lookup array?



Are you looking for an **exact, approximate, or wildcard** match?



Do you want to search **top down** or **bottom up**?

IMPORTANT NOTE: XLOOKUP is currently only available for **Office 365** subscribers

XLOOKUP

- ✓ Can retrieve a **dynamic array** of results
- ✓ Can lookup values **anywhere** in an array (left or right, horizontal or vertical)
- ✓ Defaults to **exact match**
- ✓ Supports native **wildcard** text matching
- ✓ Includes **built-in error handling** when a lookup value is not found
- ✓ Can find approximate matches in **unsorted** lists
- ✓ Can search **top-down** or **bottom-up**

VLOOKUP

- ✗ Can only return a **single value**
- ✗ Can only lookup values to the **right**, requires **HLOOKUP** for horizontal matching
- ✗ Defaults to **approximate match**
- ✗ Does not natively support **wildcard** matching
- ✗ Requires an additional **IFERROR** function for error handling
- ✗ Requires **sorted** lists for approximate matching
- ✗ Only searches **top-down**

The **CHOOSE** function selects a value, cell reference, or function to perform from a list, based on a given index number

=CHOOSE(index_num, value1, [value2], ...)



Which item in the following list should be evaluated?

1st item in the list

2nd item in the list

3rd, 4th, 5th, etc...

FUN FACTS ABOUT CHOOSE:

- List items can include **numbers, cell references, defined names, formulas, or text** (*or a mix!*)
- CHOOSE acts like an **INDIRECT** function, and can interpret cell references instead of treating them as text
- You can combine CHOOSE with other functions, or nest it directly into a cell reference

The **OFFSET** function is similar to **INDEX**, but can return either the value of a cell within an array (like INDEX) or a specific *range* of cells

=OFFSET(reference, rows, columns, [height], [width])

What's your
starting
point?

How many
rows down
should you
move?

How many
columns over
should you
move?

If you want to return a
multidimensional array,
how tall and wide should it
be?

An **OFFSET** formula where [height]=1 and [width]=1 will operate exactly like an INDEX. A more common use of **OFFSET** is to create dynamic arrays (like the Scroll Chart example in the appendix)



PRO TIP:

Don't use OFFSET or INDEX/MATCH when a simple VLOOKUP will do the trick



TEXT FUNCTIONS

Text functions can be used to standardize formatting, particularly the **TRIM**, **UPPER**, **LOWER**, and **PROPER** functions:

	A	B	C	D
1	Sample Text String	Formula	Output	Notes
2	SAMPLE sentence	=TRIM(A2)	SAMPLE sentence	<i>Removes any leading or trailing spaces from a text string</i>
3	SAMPLE sentence	=LOWER(A3)	sample sentence	<i>Converts all characters in a text string to lower case</i>
4	SAMPLE sentence	=UPPER(A4)	SAMPLE SENTENCE	<i>Converts all characters in a text string to upper case</i>
5	SAMPLE sentence	=PROPER(A5)	Sample Sentence	<i>Converts all characters in a text string to proper case (first letter capitalized)</i>
6				



PRO TIP:

If two text strings are identical except one has a trailing space, they will look exactly the same but Excel will treat them as completely different values; TRIM will make them equivalent

CONCATENATE allows you to combine text, cell values, or formula outputs into a single text string

Note: Rather than typing "**=CONCATENATE**(*Text1, Text2...*)", you can simply separate each piece of the resulting text string with an ampersand ("**&**")

	A	B	C	D
1	First Name	Last Name	Formula	Output
2	Daniel	Wright	=A2&B2	DanielWright
3	Daniel	Wright	=A3&" "&B3	Daniel Wright
4	Daniel	Wright	=LEFT(A4,3)&" "&B4	Dan Wright
5	Daniel	Wright	=LEFT(A5,3)&" "&LEFT(B5,1)&"."	Dan W.

The **LEFT**, **MID**, and **RIGHT** functions return a specific number of characters from a location within a text string, and **LEN** returns the total number of characters

=LEFT(text, [num_chars])

=RIGHT(text, [num_chars])

=MID(text, start_num, num_chars)

	A	B	C	D
1	Sample Text String	Formula	Output	Notes
3	MA-02215%AAA%_100	=LEFT(A3,2)	MA	Returns 2 characters, starting from the left
5	MA-02215%AAA%_100	=MID(A5,4,5)	02215	Returns 5 characters from the middle of the string, starting with position 4
7	MA-02215%AAA%_100	=RIGHT(A7,3)	100	Returns 3 characters, starting from the right
9	MA-02215%AAA%_100	=LEN(A9)	17	Returns the length of the string (=17 characters)

The **TEXT** function converts a numeric value to text and assigns a particular format

=TEXT(value, format_text)

Numeric value, formula that evaluates to a numeric value, or reference to a cell containing a numeric value

Numeric format as a text string enclosed in quotes (i.e. "m/d/yyyy", "\$0.00" or "###0.00")

	A	B
1	Name	Earnings
2	Tim	\$4,500
3	George	\$3,250
4	Lisa	\$3,725

= "Lisa earned "&B4 *returns* "Lisa earned 3725"

= "Lisa earned "&TEXT(B4"\$#,###") *returns* "Lisa earned \$3,725"



PRO TIP:

Use **VALUE** to convert a text string that represents a number into a value

The **SEARCH** function returns the number of the character at which a specific character or text string is first found (otherwise returns #VALUE! error)

=SEARCH(find_text, within_text, [start_num])

What character or string
are you searching for?

Where is the text that
you're searching
through?

Search from the beginning (default)
or after a certain number of
characters?

	A	B	C	D
11	MA-02215%AAA%_100	=SEARCH("%",A11)	9	Searches the string for "%" and returns the position
13	MA-02215%AAA%_100	=SEARCH("%",A13,10)	13	Searches for "%", starting with the 10th character, and returns the position
15	MA-02215%AAA%_100	=MID(A15,SEARCH("%",A15),5)	%AAA%	Returns 5 chars from the middle of the string, beginning where it finds the "%"
17	MA-02215%AAA%_100	=MID(A17,SEARCH("%",A17)+1,3)	AAA	Returns 3 characters from the middle of the string, beginning 1 position after "%"



PRO TIP:

The **FIND** function works exactly the same way, but is case-sensitive

IF(ISNUMBER(SEARCH is powerful combination of functions that can be used to classify data based on cells that contain specific strings of text

=IF(ISNUMBER(SEARCH(find_text, within_text)),value_if_true, value_if_false)

Searches for a specific string of text within a given cell

Returns one value if that string is found (TRUE), and another if it is not found (FALSE)

	A	B
1	Placement	Media
2	12983-Aff-160x90_small	Other
3	982308-Disp-160x90_large	Display
4	23124-Aff-160x90_small	Other
5	463-Disp-160x90_small	Display
6	390238-Agg-160x90_large	Other

=IF(ISNUMBER(SEARCH("Disp",A2)),"Display","Other")

Search the cells in column A for the text string "Disp" and classify column B as "Display" if you find it, "Other" if you don't

The background of the banner is a dark green overlay on a blurred image of an Excel spreadsheet. The spreadsheet contains various financial data points, including dates like 11/2/2014, 11/9/2014, 11/16/2014, 11/23/2014, and 11/30/2014. It also features numerical values, percentages, and some faint line and bar charts. The overall aesthetic is professional and data-oriented.

DATE & TIME FUNCTIONS

Every date in Excel has an associated **date value**, which is how Excel calculates the passage of time (using midnight on 1/1/1900 as the starting point)

Excel recognizes most typed dates and automatically applies a common format (i.e. m/d/yyyy), along with an associated date value (cell format → General)

***Note:** If you type a date in a format that Excel does NOT recognize, it will be treated as text and there will be no associated date value; however, you can use a **DATEVALUE** or **TIMEVALUE** function to convert unformatted dates or times into serial values*

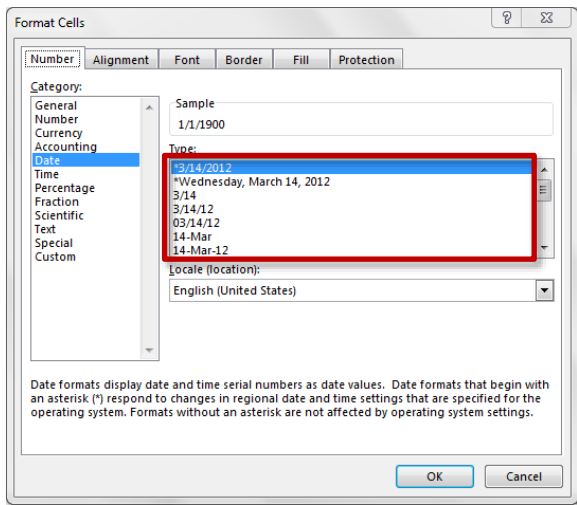
Date	Date Value
1/1/1900	1
1/11/1900	11
2/6/2015	42041
2/6/15 12:00 PM	42041.5
2/6/15 6:00 PM	42041.75

Jan 1, 1900 is the first date with an assigned date value (1). Feb 6, 2015 is the 42,041st day since 1/1/1900, so its date value = 42041

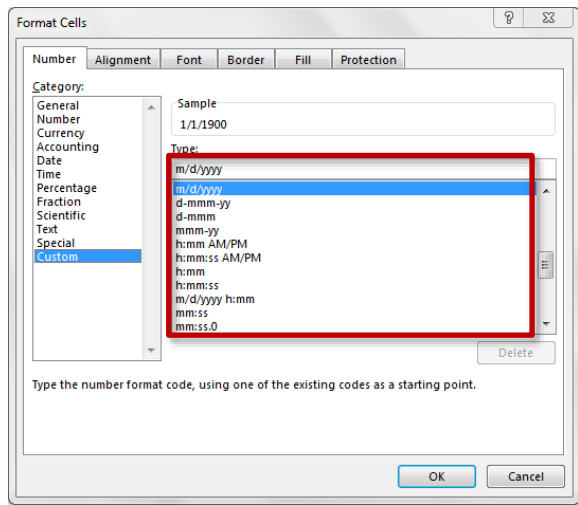
Date values can also indicate fractions of days: 42041.5 translates to noon on 2/6/2015 (50% through the day), and 42041.75 translates to 6:00pm on 2/6/2015 (75% through the day)

To format dates in Excel, you can either select a preset option from the “Date” category of the “Format Cells” dialog box, OR create your own **custom format**

Preset Formats:



Custom Format:

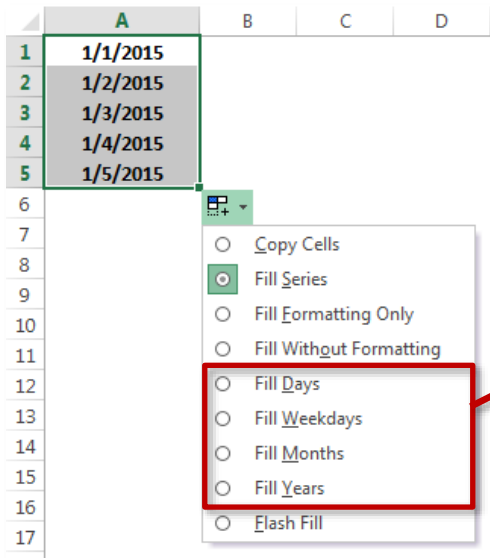


You can build your own custom formats using combinations of date/time codes. For example:

d = day w/out leading zero (1-31)
dd = day w/ leading zero (01-31)
ddd = day-of-week (Sat)
dddd = day-of-week (Saturday)
m = month w/out leading zero (1-15)
mm = month w/ leading zero (01-15)
mmm = month abbreviation (Jan)
mmmm = full month (January)
yy = last 2 digits of year (15)
yyyy = full year (2015)

(full list available at support.office.com)

When you drag the corner of a cell containing a date, Excel automatically applies subsequent values automatically using **Fill Series** options:



*Click the **Auto Fill Options** button to determine exactly which values your subsequent cells should take:*

Copy Cells = Repeats the same value in all cells

Fill Days = Increases the date by 1 day per cell

Fill Weekdays = Increases the date by 1 day per cell (excluding weekends)

Fill Months = Increases the date by 1 month per cell

Fill Years = Increases the date by 1 year per cell

The **TODAY()** and **NOW()** functions return the current date or exact time

***Note:** These are **volatile** functions, meaning that they change with every worksheet calculation*

TODAY()=

2/6/2015

NOW()=

2/6/2015 17:15

*This is what the **TODAY()** and **NOW()** functions return at 5:15pm on February 6, 2015. Note that these values will automatically update with every change made to the workbook*



PRO TIP:

*Make sure to enter **TODAY()** and **NOW()** functions with both parentheses included – these functions don't refer to other cells*

Excel will always calculate dates and times based on their *precise* underlying serial values, but what if you need to work with less-specific values, like months instead of days, or hours instead of seconds?

The **YEAR**, **MONTH**, **DAY**, **HOUR**, **MINUTE**, and **SECOND** functions extract individual components of a given date:

	A	B	C	D	E	F	G
1		YEAR	MONTH	DAY	HOUR	MINUTE	SECOND
2	2/6/2015 17:57	2015	2	6	17	57	16
3		=YEAR(A2)	=MONTH(A2)	=DAY(A2)	=HOUR(A2)	=MINUTE(A2)	=SECOND(A2)
4							

Use the **EOMONTH** function to calculate the last day of a given month, or to calculate the start/end dates of previous or future months

=EOMONTH(start_date, months)

Reference to the cell containing
the start/current date

Number of months before or after the start/current date (positive
number yields a date in the future, negative number yields a date in
the past)

	A	B	C
1			
2		Current Date:	8/3/2015
3			
4		End of month:	8/31/2015
5		Start of Month:	8/1/2015
6		Start of Next Month:	9/1/2015

→ =EOMONTH(C2, 0)

→ =EOMONTH(C2, -1)+1

→ =EOMONTH(C2, 0)+1

YEARFRAC calculates the fraction of a year represented by the number of whole days between two dates

=YEARFRAC(start_date, end_date, [basis])

Reference to the cell
containing the start date

Reference to the cell
containing the end
date

Option specify the type of day count to use:

- 0 (default)** = US (NASD) 30/360
- 1** = actual/actual (**RECOMMENDED**)
- 2** = actual/360
- 3** = actual/365
- 4** = European 30/360

	A	B
1		
2	Start Date:	1/1/2015
3	End Date:	2/28/2015

=YEARFRAC(B2, B3, 1) = 15.9%
=YEARFRAC(B2, B3, 2) = 16.1%



PRO TIP:

YEARFRAC is a great tool for pacing and projection calculations

If you want to know which day of the week a given date falls on, there are two ways to do it:

1) Use a custom cell format of either “ddd” (Sat) or “dddd” (Saturday)

*-Note that this doesn't change the underlying **value**, only how that value is displayed*

2) Use the **WEEKDAY** function to return a serial value corresponding to a particular day of the week (either 1-7 or 0-6)

=WEEKDAY(serial_number, [return type])



This refers to a cell containing a **date** or **time**



0 (default) = Sunday (1) to Saturday (7)

1 = Monday (1) to Sunday (7)

3 = Monday (0) to Sunday (6)

WORKDAY returns a date that is a specified number of days before or after a given start date, excluding weekends and (optionally) holidays; **NETWORKDAYS** counts the number of workdays between two dates:

=WORKDAY(start_date, days, [holidays])

This refers to the cell containing the start date

Number of days before or after start date

Optional reference to a list of holiday dates

=NETWORKDAYS(start_date, end_date, [holidays])

This refers to the cell containing the start date

This refers to the cell containing the end date

Optional reference to a list of holiday dates

	A	B
1		
2	Start Date:	1/1/2015
3	End Date:	2/28/2015

=WORKDAY(B2, 20) = 1/29/2015

=NETWORKDAYS(B2, B3) = 42

DATEDIF calculates the number of days, months, or years between two dates

=DATEDIF(start_date, end_date, unit)

Reference to the cell
containing the start date

Reference to the cell
containing the end
date

How do you want to calculate the difference?

"D" = # of days between dates

"M" = # of months between dates

"Y" = # of years between dates

"MD" = # of days between dates, ignoring months and years

"YD" = # of days between dates, ignoring years

"YM" = # of months between dates, ignoring days and years

	A	B
1		
2	Start Date:	1/1/2015
3	End Date:	2/28/2015

=DATEDIF(B2, B3, "D") = 58

=DATEDIF(B2, B3, "MD") = 27

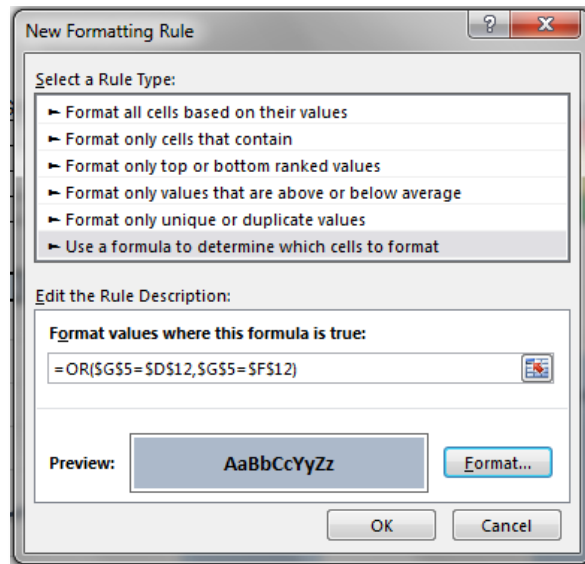
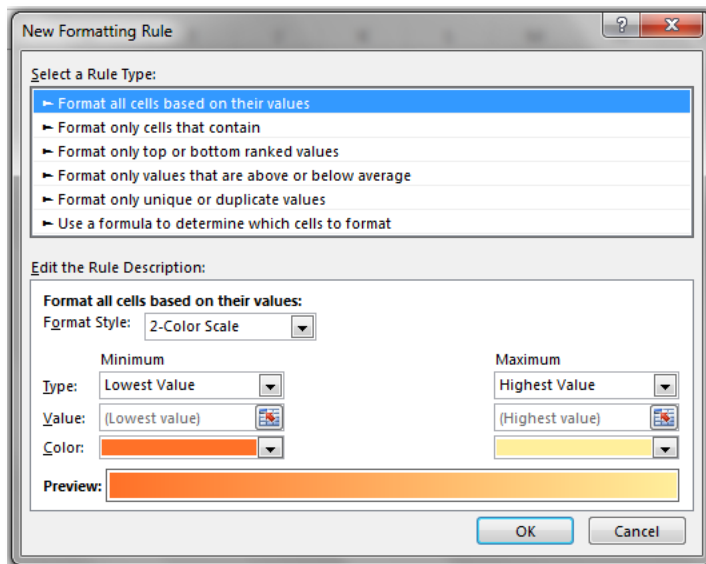
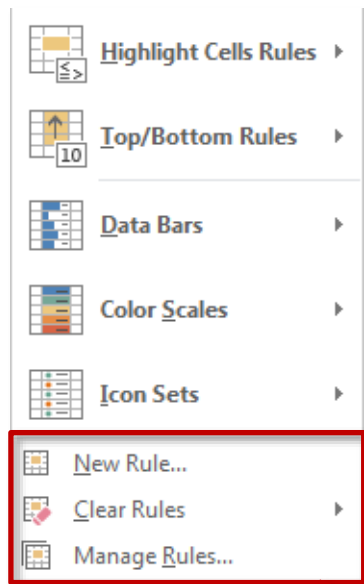


PRO TIP:

If you only need to calculate the # of days between dates, just use subtraction

FORMULA-BASED FORMATS

If you want to go rogue, you can adjust the style of existing conditional formats or create your own **formula-based rules**



This is where you can add, clear, and manage your conditional formatting rules

Formula-Based Formatting

State:						
State		Population	Student Pop.	SAT Participation Rate	Mean Verbal Score	Mean Math Score
Alabama		4447100	177884	9%	559	554
Alaska		626932	12539	51%		
Arizona		5130632	102613	34%		
Arkansas	2001	2673400	53468	6%		
California	2001	33871648	1016149	51%		
Colorado	2001	4301261	172050	31%		
Connecticut	2001	3405565	102167	82%		
D.C.	2001	783600	23508	56%		
Delaware	2001	572059	17162	67%		

In this example we're formatting the cells in columns B through H with a green fill and bold text, but only when the state name is equal to the value in cell \$C\$2

Note that the row label is relative (no "\$"), which allows us to apply this formatting to other rows without losing functionality

Edit Formatting Rule

Select a Rule Type:

- Format all cells based on their values
- Format only cells that contain
- Format only top or bottom ranked values
- Format only values that are above or below average
- Format only unique or duplicate values
- Use a formula to determine which cells to format

Edit the Rule Description:

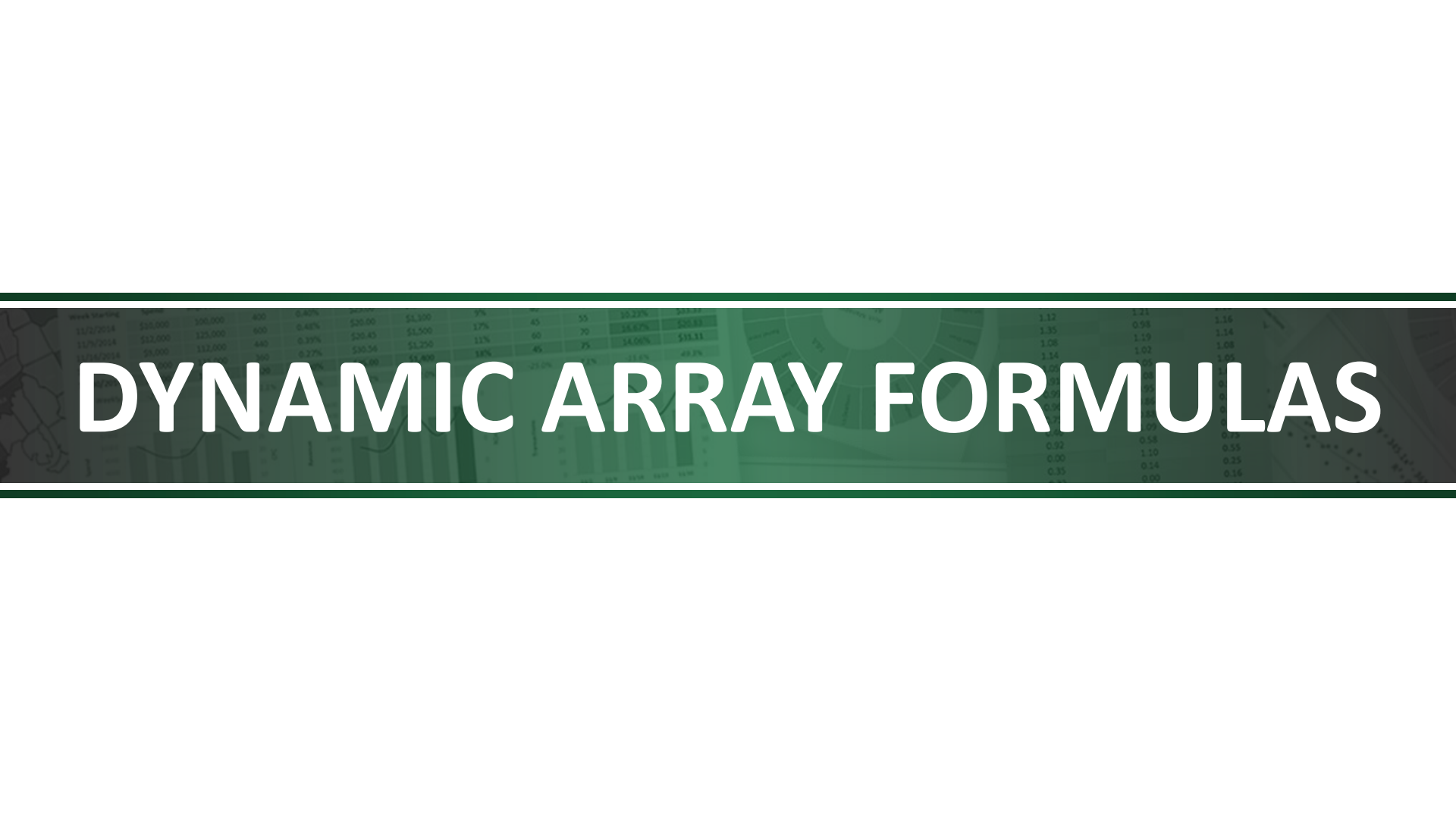
Format values where this formula is true:

=B6:H6=\$C\$2

Preview: AaBbCcYyZz

Format...

OK Cancel

The background of the slide is a dark green banner with a faint, semi-transparent image of an Excel spreadsheet. The spreadsheet contains various financial data points, including dates like '11/2/2014', '11/9/2014', and '11/16/2014', as well as numerical values and percentages. There are also some faint line and bar charts visible in the background.

DYNAMIC ARRAY FORMULAS